Sukhrob Ilyosbekov

Computer Vision · Deep Learning · Explainable AI

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Research Interests

Making vision models trustworthy enough to deploy. My work runs along three threads: interpretability a clinician can audit rather than take on faith, control over black-box generative systems from outside the model, and representation learning that transfers to scientific imaging. Building ML for a HIPAA-regulated clinical platform is where those same questions arrive with consequences attached.

Education

Northeastern University, Boston, MA *January 2024 – May 2026*

- Master of Science in Computer Science; GPA: 4.0/4.0
- Relevant Coursework: CS 7180 Advanced Perception, Machine Learning, Computer Vision

Suffolk University, Boston, MA *September 2021 – May 2023*

- Bachelor of Science in Computer Science; Cum Laude; GPA: 3.5/4.0
- Transferred from INTI International College, Malaysia (2019 – 2021) and Tashkent University of Information Technologies, Uzbekistan (2017 – 2019)

Publications

[1] **S. Ilyosbekov**, S. Gajjar, and R. Jin, "MorphoCLIP: Text-Supervised Contrastive Learning for Perturbation Matching in Cell Painting Images," *arXiv preprint*, arXiv:2608.22690, 2026. [Paper]

[2] **S. Ilyosbekov**, "Localize, Don't Beautify: Client-Side Control of Image-Editing APIs for Cosmetic Surgery Previews," *arXiv preprint*, arXiv:2608.02841, 2026. [Paper]

[3] **S. Ilyosbekov**, "MelanomaNet: Explainable Deep Learning for Multi-Class Skin Lesion Classification," *arXiv preprint*, arXiv:2512.09289, 2025. [Paper]

Awards & Honors

- **National Physics Olympiad, 1st Place,** Uzbekistan; first student from my high school to win nationally
- **Associate Member,** Sigma Xi, The Scientific Research Honor Society (2026)
- **University Coding Competition Winner,** Tashkent University of Information Technologies
- **Outstanding Project Award,** Bookshelf Scanner, Computer Vision Course, Northeastern University

Research

MorphoCLIP: Text-Supervised Cell Painting Retrieval GitHub | Paper

- Built a text-supervised contrastive model that matches Cell Painting microscopy images of drug- and gene-perturbed cells to natural-language treatment descriptions, pairing frozen vision and language backbones (DINOv3, BioClinical ModernBERT) with trainable projection heads so the whole model trains on a single consumer GPU.
- Evaluated bidirectional image-to-text and text-to-image retrieval on the CPJUMP1 benchmark (51 plates, 3M+ cell images, 303 drugs, 160 genes), with batch correction in the embedding space to remove plate-to-plate variation.
- Collaborative work with Northeastern co-authors S. Gajjar and R. Jin; I led the model design, training, and evaluation.
- **Stack:** PyTorch, DINOv3, BioClinical ModernBERT, contrastive learning, CPJUMP1

Localize, Don't Beautify: Client-Side Control of Image-Editing APIs Paper

- Asked how much of a black-box editor's behavior can be controlled from outside the model, with cosmetic-surgery previews as the test case: commercial APIs beautify the whole face when asked to change one feature.
- Compared prompt-only, masked-compositing, and model-based inpainting across six commercial editors and one inpainting model on 196 facelift and rhinoplasty edits, scoring identity preservation (ArcFace) and localization accuracy. Masked compositing beat model-based inpainting on localization.
- **Stack:** Image-editing APIs, inpainting models, ArcFace, prompt engineering

MelanomaNet: Explainable Deep Learning for Skin Lesion Classification GitHub | Paper

- Trained an EfficientNet V2 classifier across all 9 ISIC 2019 diagnostic categories at 384×384 resolution, reaching 85.6% accuracy and 0.856 weighted F1 on 25,331 dermoscopic images, using focal loss against the heavy class imbalance.

- Decomposed GradCAM++ attention along the ABCDE criteria dermatologists already use, quantifying asymmetry, border irregularity, color variation (K-means), and diameter directly from the lesion mask.
- Designed alignment metrics between model attention and the extracted clinical features, so the interpretability claim is measured rather than argued from a handful of example heatmaps.
- **Stack:** PyTorch, EfficientNet V2, GradCAM++, OpenCV, focal loss, ISIC 2019

Selected Course Projects

LightDepth: Lightweight Monocular Depth Estimation — GitHub
- Designed a ResNet18 encoder with a U-Net decoder and skip connections for monocular depth, holding accuracy with 42% fewer parameters (14.3M vs 24.8M) than Depth Anything V2.
- Ran 72% faster at inference while improving relative error on NYU Depth V2 (5.3% better AbsRel, 31.1% better SqRel).
- **Stack:** PyTorch, ResNet18, U-Net, NYU Depth V2

FSRCNN: Accelerating Super-Resolution Convolutional Neural Network — GitHub
- Reproduced FSRCNN (Dong et al., ECCV 2016) for single-image super-resolution at 2×, 3×, and 4× scales, with end-to-end learned upsampling giving a 40× speedup over SRCNN.
- Matched the reported gains on Set5 (+1.78 dB PSNR, +0.0507 SSIM) and Set14 (+1.26 dB PSNR, +0.0477 SSIM), with ablations on the shrinking and mapping layers.
- **Stack:** PyTorch, mixed-precision training, T91/Set5/Set14/DIV2K

Bookshelf Scanner: Multi-Modal Book Detection and Recognition — GitHub
- Combined YOLO instance segmentation to isolate each book spine on a shelf with the Moondream2 vision-language model to read title and author off each spine, an end-to-end detect-then-ground pipeline. Received the course Outstanding Project Award.
- **Stack:** YOLO, Moondream2 VLM, llama.cpp, PyTorch, FastAPI

Work Experience

Senior AI/ML Engineer | EmTech Care Labs | Portland, ME — *Jan 2025 – Present*
- Built the machine-learning features for a HIPAA-compliant Alzheimer's care platform, including a decline-risk model over daily activity and care-log data for roughly 1,200 residents, reaching 0.81 AUC and surfacing early signs about three weeks before the care team escalated on its own.
- Added an NLP pipeline that reads free-text clinical notes and extracts structured details (symptoms, medications, care events) at roughly 0.90 F1 on a held-out set, and stood up model serving and monitoring on de-identified data under HIPAA constraints.

Machine Learning Engineer | AllFactors | Santa Clara, CA — *Oct 2021 – Dec 2024*
- Built property-valuation and market-forecasting models over roughly 2M listings across 40+ metros, cutting median absolute valuation error from about 9% to 5% against the rules-based estimate the product shipped with.
- Designed the real-time feature pipeline that fed live market data to the models and pushed updated predictions to users, then productionized the models into the customer-facing dashboards.

Earlier: Software Engineer at Smart Meal Service, Russia (2020 – 2021), control software for self-service food kiosks. Game Developer at Pentalight Technology, Malaysia (2020 – 2021), multiplayer networking and hand tracking for a VR smart-city simulation. Freelance Developer (2019 – 2020), Python NLP and .NET applications. Game Developer at EC Dev Team, Uzbekistan (2016 – 2019).

Technical Skills

Deep Learning & CV: PyTorch, TensorFlow/Keras, OpenCV, CNNs, Vision Transformers, U-Net, ResNet, EfficientNet, YOLO, contrastive and self-supervised learning, GradCAM++, classification, detection and segmentation, depth estimation, super-resolution, explainable AI

Mathematics: Linear algebra, probability and statistics, convex optimization, numerical methods, information theory, differential equations, hypothesis testing and experimental design

Machine Learning: Supervised and unsupervised learning, transfer learning, loss design, regularization, hyperparameter search, cross-validation and error analysis, XGBoost, K-means, NLP

Research Computing: Python, NumPy/SciPy, pandas, scikit-learn, Hugging Face, CUDA, mixed-precision and multi-GPU training, Weights & Biases, Docker, LaTeX

Software Engineering: C#, C++, TypeScript; .NET, FastAPI, PostgreSQL, Kubernetes, AWS, Azure